CYBERGUARD-X — AI SOC THREAT INTELLIGENCE PLATFORM

Forensic Architecture & Feature Upgrade Technical Report

Before vs. After Analysis (URL, Image, and Video Intelligence Engines)

1. URL Intelligence & Phishing Telemetry Engine

Category	Earlier Capability (Before)	Upgraded Advanced Capability (After)
Live DNS & Host Verification	Checked only static string parameters. Could not verify if the domain actually exists on the Internet.	**Socket-Level DNS Resolution:** Performs real-time IPv4 resolution. Detects dead, unconfigured, or NXDOMAIN hosts (e.g. *https://nishat.com*) immediately.
Brand Spoofing & Impersonation	Simple substring check against static strings.	**Levenshtein & Homoglyph Engine:** Computes character distance against 20+ top brands (PayPal, Netflix, Microsoft, Google, SBI, Apple) to catch subtle typosquats.
DGA & Algorithmic Domains	No mathematical or algorithmic domain inspection.	**Shannon Entropy Analysis:** Measures domain name character entropy (bits). Detects machine-generated botnet DGA domains.
TLD Risk & Abuse Tiering	Only checked standard HTTP vs HTTPS protocol.	**TLD Abuse Classification:** Evaluates high-risk abusive TLDs (.tk, .top, .xyz, .click, .ru, .pw, .fit) and assigns weighted risk multipliers.
SOC Explainability & UI	Displayed generic risk score with brief text.	**Dedicated DNS Forensics Panel:** Displays resolved IP, Host Type, Entropy bits, TLD Risk Tier, and Brand Target details.

2. Image Authenticity & Content Safety Forensics

Category	Earlier Capability (Before)	Upgraded Advanced Capability (After)
Adult / NSFW Content Detection	**Zero Content Safety:** An unedited nude photo scored only 25/100 (Likely Authentic) because camera noise was natural.	**AI Anatomical Skin Exposure Engine:** Dual-color space (HSV + YCrCb) skin segmentation + contour density. Over 30% unclad skin triggers **98% Critical Risk** & Policy Violation.
Steganography & Trailing Payloads	No file structure or end-of-frame inspection.	**EOF Boundary Dissection:** Scans past JPEG FF D9 and PNG IEND markers. Detects appended ZIP archives, PowerShell scripts, and PE (.exe) binaries with **94% Critical Risk**.
EXIF Metadata & Tampering	Camera firmware was occasionally misflagged as photo editing software.	**Smart Provenance Engine:** Distinguishes legitimate camera firmware from software editing suites (Photoshop, GIMP, Canva, Lightroom).
1-Click Test Scenarios	Limited static samples without specific attack scenarios.	**5 Interactive Scenarios:** Explicit Adult Demo, Steganography Payload Demo, AI Deepfake, Altered ID, and Authentic Media.
SOC UI Telemetry	Displayed only basic manipulation percentage.	**Content Safety & Stego Cards:** Shows Skin Exposure %, Policy Threshold %, Largest Patch %, and Embedded Payload byte size.

3. Video Stream & Deepfake Forensics Engine

Category	Earlier Capability (Before)	Upgraded Advanced Capability (After)
Keyframe Sampling	Read only file metadata without processing temporal video frames.	**Multi-Keyframe Temporal Sampling:** Extracts 12 dynamic keyframes across video timeline via OpenCV for deep forensic analysis.

Deepfake Face-Swap Detection	No temporal motion flow or jitter inspection.	**Temporal Motion Flow Jitter Engine:** Measures standard deviation of frame-to-frame pixel differences (diff_std) to catch AI face-swap flickering.
Video Splicing & Cuts	Could not identify splices or video cut transitions.	**Histogram Correlation (min_corr):** Detects abrupt color distribution drops indicating digital scene seams or spliced clips.
Optical Blur Stability	No temporal sharpness variance analysis.	**Laplacian Sharpness Tracking:** Analyzes intra-frame blur stability to detect selective facial blurring artifacts.
1-Click Video Demos & Telemetry	Placeholder state with no live sample files.	**Interactive Video Panel:** Integrated with AI Deepfake Video, Spliced Clip, and Authentic Video with comprehensive live metrics.

EXECUTIVE CONCLUSION: CyberGuard-X has evolved from a basic static-checking tool into an enterprise-grade **Multi-Vector AI Threat Defense Platform**. By combining socket DNS lookups, Shannon entropy, dual HSV/YCrCb anatomical skin segmentation, EOF file boundary steganography scanning, and temporal OpenCV motion flow heuristics, the platform delivers zero-trust security intelligence with verifiable forensic proofs.